

FORENSIC ANALYSIS REPORT

Multi-Vendor DVR/NVR Forensic Analysis Tool

Case Number:	CASE-2026-BLR-089	Report Date:	01-10-2026 22:37
Case Title:	Cyber Hub Data Center & Vault Room Intrusion	Status:	ACTIVE
Officer:	Insp. Rajesh Varma, Cyber Forensic Division	Organization:	State Cyber Crime Investigation Cell (SCCIC)

1. EVIDENCE ITEMS

1.1 Evidence #EV-2026-HK-001

Device Type	Hikvision Network Video Recorder (NVR)
Vendor	Hikvision
Model	DS-7616NI-I2/16P (16-Ch Embedded 4K)
Serial Number	HK-SN-998274102-X
Acquisition Method	Tableau T8u Hardware Write-Blocker (Bit-Stream DD Image)
Original SHA-256	41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d
Verified SHA-256	41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d
Integrity Status	VERIFIED
Acquired By	Insp. Rajesh Varma
Acquired At	2026-09-28 18:57:07

1.2 Evidence #EV-2026-DH-002

Device Type	Dahua 8-Channel NVR Storage Disk
Vendor	Dahua
Model	DHI-NVR5208-8P-4KS2E
Serial Number	DH-SN-33918274-B
Acquisition Method	Atola Insight Forensic Imager (E01 Format)
Original SHA-256	c4ceefc6e39102fb76773e7d3ba2662c1bb47af2e641102eaa3ca08bfc18cc4d
Verified SHA-256	c4ceefc6e39102fb76773e7d3ba2662c1bb47af2e641102eaa3ca08bfc18cc4d
Integrity Status	VERIFIED
Acquired By	Insp. Rajesh Varma
Acquired At	2026-09-29 02:57:07

2. RECOVERED VIDEO FILES

#	Filename	Codec	Resolution	Duration	Channel	Status
1	CH01_Vault_Entrance_0200_0330.	H.264 / AVC (High Profile)	1920x1080 (Full HD)	5400.0s	1	TAMPER_DETECTED
2	CARVED_CH02_Deleted_Sector_0x4	H.264 / AVC (Main Profile)	1920x1080 (Full HD)	1820.0s	2	VALID
3	CH04_Perimeter_Security_Gate_0	H.265 / HEVC (Main 10 Profile)	3840x2160 (4K UHD)	10800.0s	4	VALID

3. TAMPER DETECTION ANALYSIS

Analysis Type: timestamp_discontinuity

Severity: HIGH

Description: Critical 4 min 12 sec (252s) timestamp discontinuity identified during the intrusion window (02:43:18 to 02:47:30 UTC). 7,560 expected video frames missing with no power-down signal recorded in syslog.

Analysis Type: gop_break

Severity: HIGH

Description: GOP (Group of Pictures) boundary violation at byte offset 0x12A9E00. Sequence terminated mid-GOP on a predictive P-frame without closed IDR boundary; next frame begins with new non-sequential sequence parameter set (SPS).

Analysis Type: watermark_integrity

Severity: MEDIUM

Description: Hikvision proprietary digital watermark check failed on 14 consecutive frames following timestamp resumption. CRC32 checksum mismatch in SEI user data unaligned.

4. CHAIN OF CUSTODY LOG

The following is a hash-chained, append-only log of all actions performed on the evidence. Each entry is cryptographically linked to the previous entry using SHA-256, ensuring that no entries can be altered, removed, or inserted without detection.

Entry #	1	Timestamp	2026-09-28 14:57:07
Action	PHYSICAL_SEIZURE	Performed By	Insp. Rajesh Varma
Entry Hash: 275f246cbfb578ef59a0509b75b4afef9bb1be1861d274d8eea76a9208559fbc			
Previous Hash: 00			
Details: Physical seizure of Hikvision NVR DS-7616NI-I2 from Data Center Rack #4 under Panchnama #72/2026. Sealed in anti-static tamper-evident evidence bag #TE-8812.			
Entry #	2	Timestamp	2026-09-28 16:57:07
Action	WRITE_BLOCKER_MOUNT	Performed By	Forensic Tech K. Sridhar
Entry Hash: dd115272beab2d692976d6f7afddabd2f791e47060cf55f502be14e3e071a2f7			
Previous Hash: 275f346cbfb578ef59a0509b75b4afef9bb1be1861d274d8eea76a9208559fbc			
Details: Seized 4TB Seagate SkyHawk HDD connected to Tableau T8u Forensic USB 3.0 Bridge in Hardware Read-Only Write-Blocker Mode. DIP switches verified.			
Entry #	3	Timestamp	2026-09-28 18:57:07
Action	BIT_STREAM_ACQUISITION	Performed By	Forensic Tech K. Sridhar
Entry Hash: 8a31c09e5824526f454d9827deff46d08f5e4b650d205a269ad7463f7e8ddf09			
Previous Hash: dd115272beab2d692976d6f7afddabd2f791e47060cf55f502be14e3e071a2f7			
Details: Raw bit-stream image (DD) completed. Size: 4,294,967,296,000 bytes. Initial SHA-256 computed: 41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d			
Entry #	4	Timestamp	2026-09-28 19:27:07
Action	HASH_VERIFICATION	Performed By	Insp. Rajesh Varma
Entry Hash: 41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d			
Previous Hash: 8a31c09e5824526f454d9827deff46d08f5e4b650d205a269ad7463f7e8ddf09			
Details: Secondary independent integrity verification passed. Computed SHA-256 matches master hash exactly: 41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d. Integrity status: VERIFIED.			
Entry #	5	Timestamp	2026-09-28 19:47:07
Action	VENDOR_DETECTION	Performed By	System Automated Engine
Entry Hash: a4d326aee3356dedafeca3b4a109b62aa8cf612ebbc5689313ecde3d6c389a77			
Previous Hash: 41b5d947f5ccb80dc3b34726c1fde14b70c83875ac69089ed6a3b35d82a6344d			
Details: Automated vendor identification executed: Hikvision proprietary HIKFS v1.2 file system detected (magic 0x48494B4653). Confidence: 99.8%.			
Entry #	6	Timestamp	2026-09-28 22:57:07
Action	DEEP_CARVING_EXECUTED	Performed By	Dr. Ananya Sen
Entry Hash: 207a0f232c7126fbd2f910855509ac958c7a53a921b6aaaff68c566eac50cd79			
Previous Hash: a4d326aee3356dedafeca3b4a109b62aa8cf612ebbc5689313ecde3d6c389a77			
Details: H.264 NAL carver executed on unallocated sector ranges 0x40000000 - 0x60000000. Successfully recovered 1 deleted video stream (Cam 02 Server Room Corridor).			
Entry #	7	Timestamp	2026-09-29 05:57:07
Action	TAMPER_AUDIT_COMPLETED	Performed By	Dr. Ananya Sen
Entry Hash: 353599a9bcbfbca0a6ad14763203a8db07ccaddf45bca47c2ff604b351a41710			
Previous Hash: 207a0f232c7126fbd2f910855509ac958c7a53a921b6aaaff68c566eac50cd79			
Details: Cryptographic and structural tamper audit concluded: 3 anomalies detected including critical 252s timestamp gap and GOP truncation at 02:43:18.			
Entry #	8	Timestamp	2026-09-30 12:57:07
Action	REPORT_GENERATION	Performed By	System Automated Engine
Entry Hash: 2f327b3a2ce003c38acc83a772deeba8ebd4516cb7b81e9ee8384f16c8dbe379			

Previous Hash: 0e96f0e809c17d2bd281b1e77957ee469adc977e4c745da90acf55ed32744826

Details: Comprehensive Court-Ready Forensic Examination Report and Certificate under Section 63 of Bharatiya Sakshya Adhiniyam, 2023 generated.

Entry #	9	Timestamp	2026-10-01 06:57:07
Action	LAB_TRANSFER	Performed By	Insp. Rajesh Varma

Entry Hash: a2ee9e8f0bf1610dddb0420bc028202fc0f2881e207e5dfc4b85754a2b222600

Previous Hash: 2f327b3a2ce003c38acc83a772deeba8ebd4516cb7b81e9ee8384f16c8dbe379

Details: Digital evidence archive transferred to Secure Evidence Vault #4 at State Cyber Crime Investigation Cell. Received by Custodian Officer D. Murthy.

CERTIFICATE UNDER SECTION 63 OF THE BHARATIYA SAKSHYA ADHINIYAM, 2023

(For admissibility of electronic records as evidence)

I, **Insp. Rajesh Varma, Cyber Forensic Division**, holding the position of Forensic Examiner at **State Cyber Crime Investigation Cell (SCCIC)**, do hereby certify the following in relation to Case Number **CASE-2026-BLR-089**:

- (a) The electronic records described herein were produced by a computer/device during the period over which the said computer/device was used regularly to store or process information for the purposes of the activities regularly carried on during that period by the person having lawful control over the use of the computer/device.
- (b) During the said period, information of the kind contained in the electronic record was regularly fed into the computer/device in the ordinary course of the said activities.
- (c) Throughout the material part of the said period, the computer/device was operating properly and that even if not operating properly or was out of operation during some part of that period, this fact did not affect the accuracy of the electronic record.
- (d) The information contained in the electronic record reproduces or is derived from such information fed into the computer/device in the ordinary course of the said activities.

Digital Evidence Hash Values (SHA-256):

Evidence #EV-2026-WK-001:	41b5d047f5eeb80dc2b24726a1fda14b70a82875ee60090ed6a2b25d82a6244d
Evidence #EV-2026-DH-002:	c4ceefc6e39102fb76773e7d3ba2662c1bb47af2e641102eaa3ca08bfc18cc4d

Signature: _____

Date: 01-10-2026

Name: Insp. Rajesh Varma, Cyber Forensic Division

Place: _____

Designation: Forensic Examiner

Organization: State Cyber Crime Investigation Cell (SCCIC)